

The spherical Earth

From last lecture

Ray Hamiltonian defined by:

$$H_k(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \|\mathbf{p}(\mathbf{x})\|^2 c_k[\mathbf{x}, \hat{\mathbf{p}}(\mathbf{x})]^2,$$

Eikonal equation:

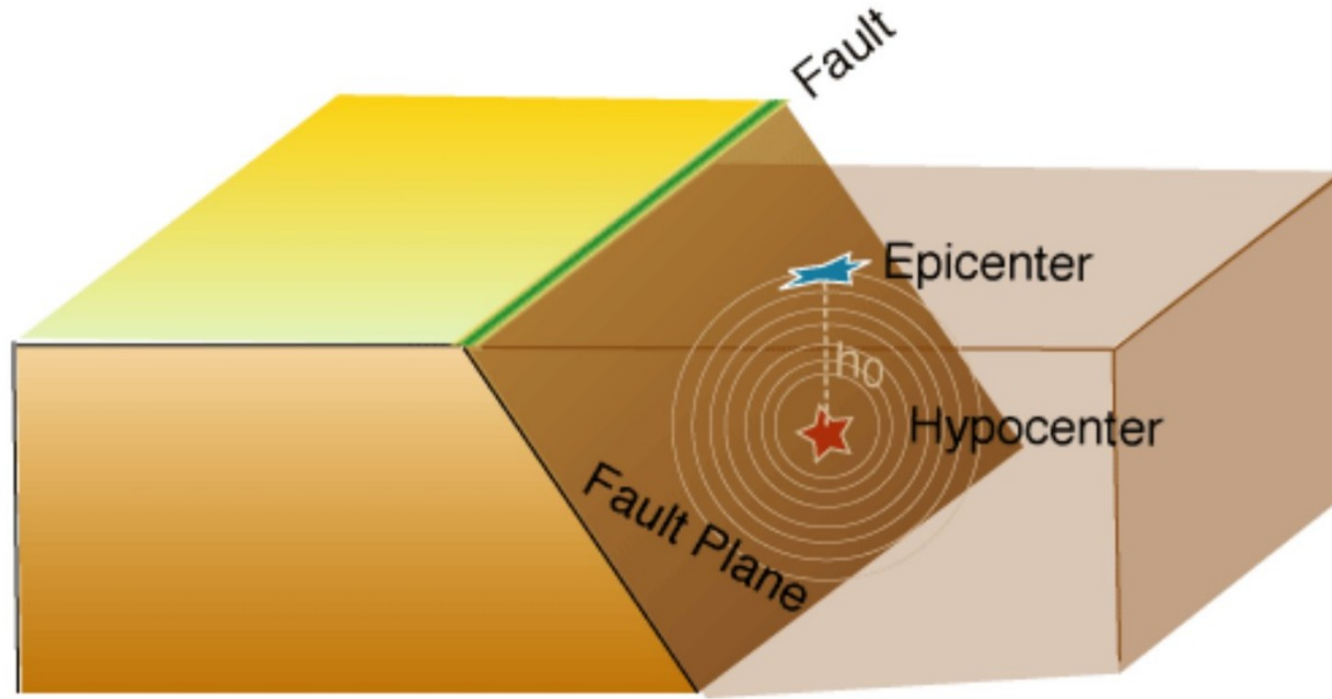
$$H_k[\mathbf{x}, \mathbf{p}(\mathbf{x})] = \frac{1}{2}, \quad \mathbf{p}(\mathbf{x}) = (\nabla T)(\mathbf{x}).$$

Solved through integration of Hamilton's equations:

$$\frac{dx_i}{d\sigma} = \frac{\partial H_k}{\partial p_i}, \quad \frac{dp_i}{d\sigma} = -\frac{\partial H_k}{\partial x_i}$$

Earthquake location

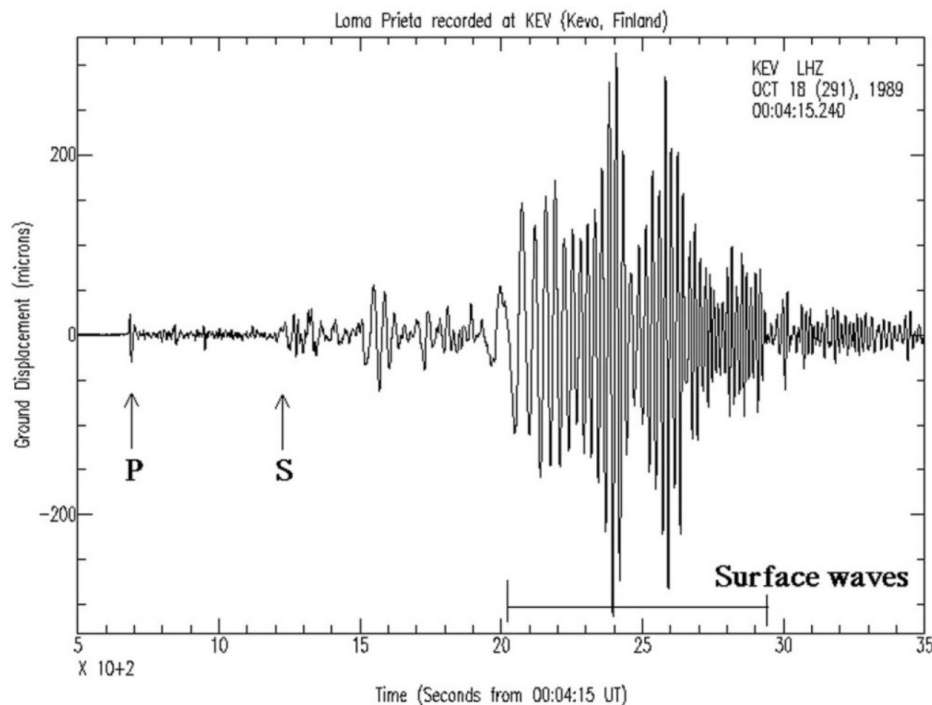
Hypocentre location



The **hypocentre** is where the rupture initiates

It is the origin of the **first waves** recorded on seismometers

Arrival time measurements



The **arrival time** is the actual time (e.g., noon on Wednesday) at which a seismic phase arrives at a seismometer. Measured from **high-frequency** recordings to allow for precise measurements

Forward modelling arrival times

Arrival time at the i th seismometer for a given phase takes the form:

$$t_i = t_h + T(\mathbf{x}_i, \mathbf{x}_h) + e_i$$

Travel time can be calculated **exactly** using ray theory. But to do this, we need to know the Earth's velocity structure

Forward problem depends on just four model parameters:

- Three components of the hypocentre position
- The hypocentral time

Inversion of arrival times

Misfit between data and proposed model parameters can be quantified:

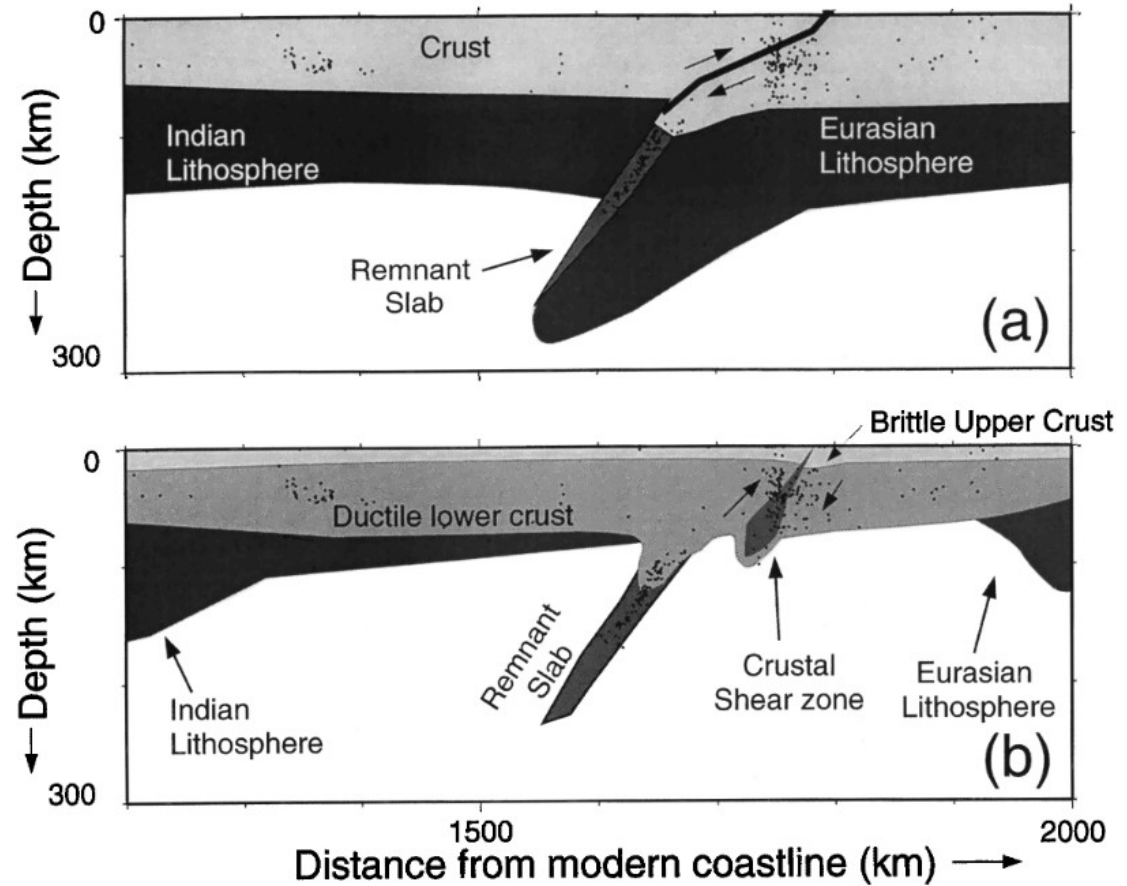
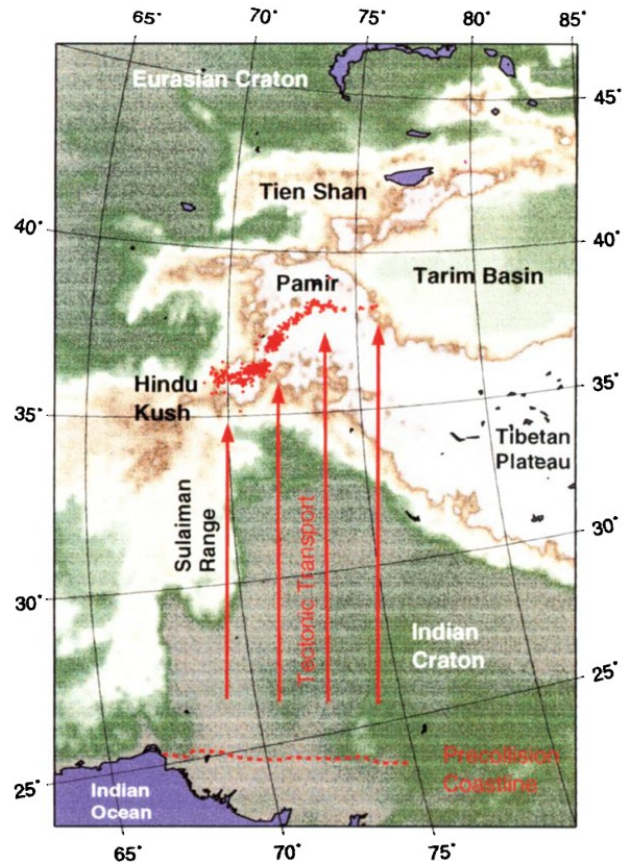
$$J(\mathbf{x}_h, t_h) = \sum_{i=1}^{N_s} \frac{[t_h + T(\mathbf{x}_i, \mathbf{x}_h) - t_i]^2}{\sigma_i^2}$$

Optimise J with respect to the four model parameters to find a solution

This can be done iteratively using gradient-based methods or through a brute-force grid search

Uncertainties can be estimated by considering a range of model parameters that fit the data sufficiently well

Accuracy ultimately limited by the **dominant wavelength** of observed waves



Application of hypocentral locations to test competing tectonic models for the formation of the Pamir-Hindu Kush mountains (from Palvis & Das 2000)

Rays in spherical Earth models

P-wave eikonal equation

We focus on an isotropic model, and look at P-waves for definiteness

Eikonal equation takes the form:

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2},$$

$$H(\mathbf{x}, \mathbf{p}) = \frac{1}{2} \alpha(\mathbf{x})^2 \|\mathbf{p}\|^2,$$

We assume further that the P-wave velocity **varies only with radius**

Hamilton's equations

General form of Hamilton's equations:

$$\frac{d\mathbf{x}}{d\sigma} = \frac{\partial H}{\partial \mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{\partial H}{\partial \mathbf{x}},$$

For the spherically symmetric P-wave problem, they become:

$$\frac{d\mathbf{x}}{d\sigma} = \alpha \hat{\mathbf{p}}, \quad \frac{d\mathbf{p}}{d\sigma} = -\frac{1}{\alpha} \frac{d\alpha}{dr} \hat{\mathbf{x}},$$

Here hats denote unit vectors

A conserved quantity

We define the vector:

$$\mathbf{q} = \mathbf{x} \times \mathbf{p}$$

Using Hamilton's equations, we find that:

$$\frac{d}{d\sigma}(\mathbf{x} \times \mathbf{p}) = \frac{d\mathbf{x}}{d\sigma} \times \mathbf{p} + \mathbf{x} \times \frac{d\mathbf{p}}{d\sigma} = \alpha \hat{\mathbf{p}} \times \mathbf{p} - \frac{1}{\alpha} \frac{d\alpha}{dr} \mathbf{x} \times \hat{\mathbf{x}} = \mathbf{0}.$$

The form of this conserved quantity follows from the rotational symmetry via Noether's theorem (but you don't have to know this!)

Reduction of the problem

Conservation of

$$\mathbf{q} = \mathbf{x} \times \mathbf{p}$$

Implies that the ray stays in the plane spanned by initial values of $\mathbf{x}$ and

Dynamics **reduced to a plane** – c.f. two body problem

Magnitude of the conserved quantity is:

$$q = \frac{r \sin \theta}{\alpha}$$

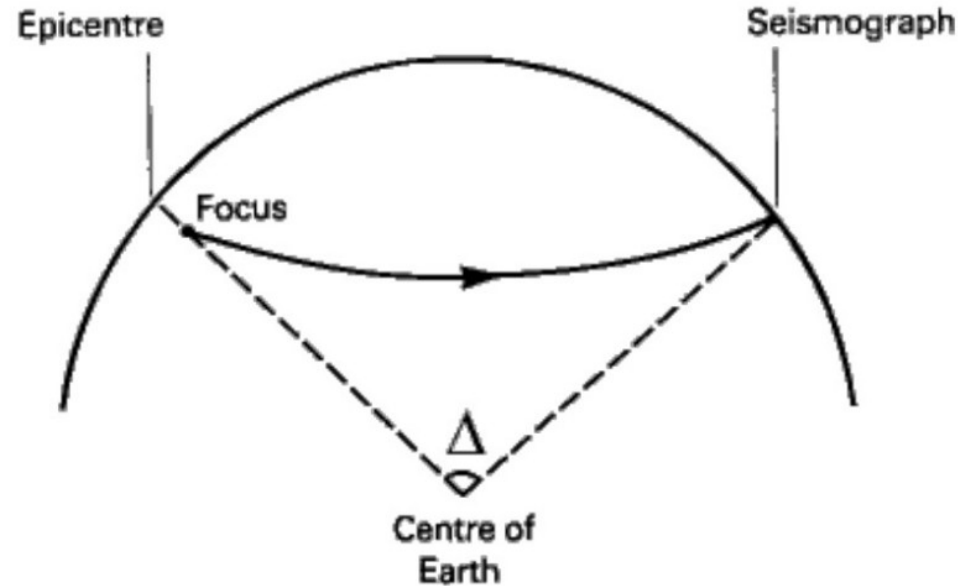
where the angle, theta, is that between $\mathbf{x}$ and $\mathbf{p}$

This is **Snell's law** expressed in spherical geometry

Reduction of the problem

Along the ray, we can write:

$$\hat{\mathbf{p}} = \cos[\theta(\mathbf{x})]\hat{\mathbf{x}} + \sin[\theta(\mathbf{x})]\hat{\Delta}(\mathbf{x}),$$



Reduction of the problem

From the conserved quantity, we can write:

$$\sin \theta = \frac{\alpha q}{r}, \quad \cos \theta = \pm \sqrt{1 - \frac{\alpha^2 q^2}{r^2}},$$

showing that the ray's direction is fixed by its radius. Plus or minus sign in the cosine term corresponds to up- or down-going rays

Using these results in Hamilton's equations, we arrive at:

$$\frac{d\mathbf{x}}{d\sigma} = -\alpha \sqrt{1 - \frac{\alpha^2 q^2}{r^2}} \hat{\mathbf{x}} + \frac{\alpha^2 q}{r} \hat{\Delta}.$$

The slowness vector has now been eliminated from the problem

Solving the reduced problem

Using the reduced ODE, we find that

$$\frac{d}{d\sigma}r[\mathbf{x}(\sigma)] = \frac{\partial r}{\partial x_i} \frac{dx_i}{d\sigma} = -\alpha \sqrt{1 - \frac{\alpha^2 q^2}{r^2}},$$

$$\frac{d}{d\sigma}\Delta[\mathbf{x}(\sigma)] = \frac{\partial \Delta}{\partial x_i} \frac{dx_i}{d\sigma} = \frac{\alpha^2 q}{r^2},$$

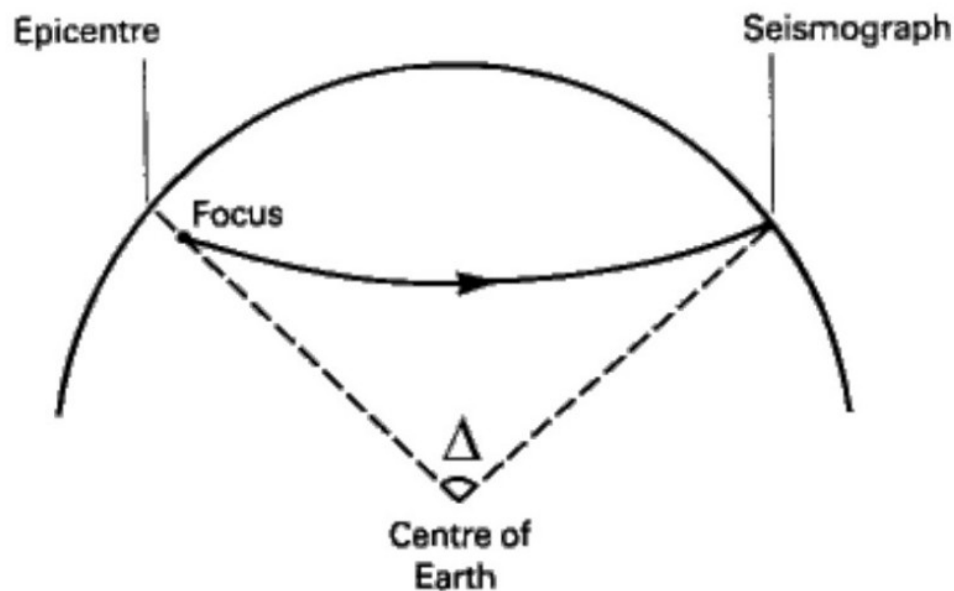
And hence, we have:

$$\frac{d\Delta}{dr} = \frac{d\Delta}{d\sigma} \left(\frac{dr}{d\sigma} \right)^{-1} = \frac{-\alpha q}{r \sqrt{r^2 - \alpha^2 q^2}}.$$

Solving the reduced problem

In a similar manner, we find:

$$\frac{dT}{dr} = \frac{dT}{d\sigma} \left(\frac{dr}{d\sigma} \right)^{-1} = \frac{-r}{\alpha \sqrt{r^2 - \alpha^2 q^2}}.$$



Solving the reduced problem

Assuming velocity decreases monotonically with radius, the ray reaches a **turning point** where it is travelling horizontally at:

$$q = \frac{r_t}{\alpha(r_t)}$$

Afterwards, it returns to the surface along a symmetric path

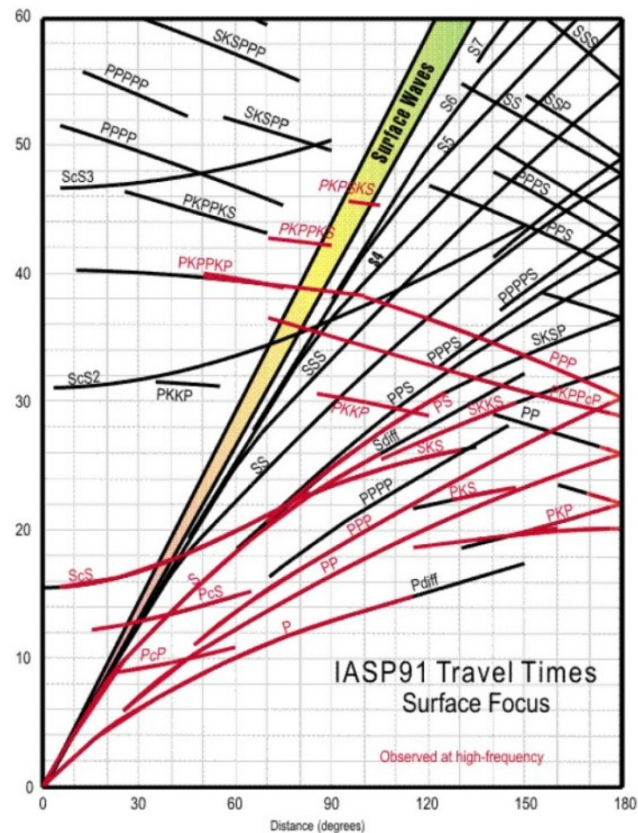
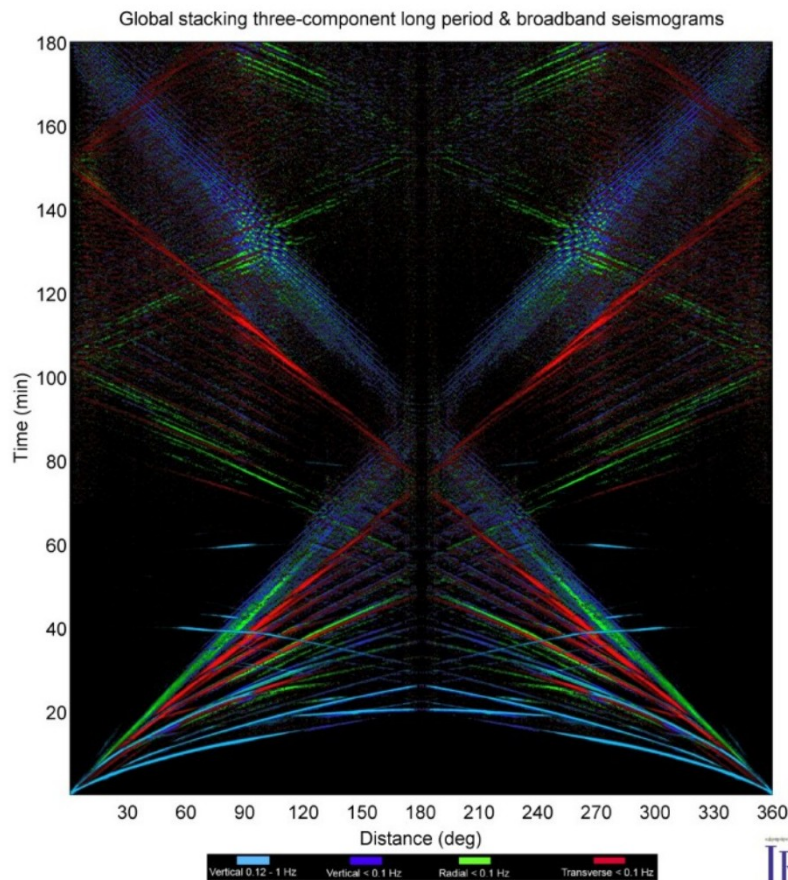
Angle and travel time expressed through the integrals:

$$\Delta = \int_{r_t}^a \frac{2\alpha(r)q}{r\sqrt{r^2 - \alpha(r)^2q^2}} dr; \quad T = \int_{r_t}^a \frac{2r}{\alpha(r)\sqrt{r^2 - \alpha(r)^2q^2}} dr.$$

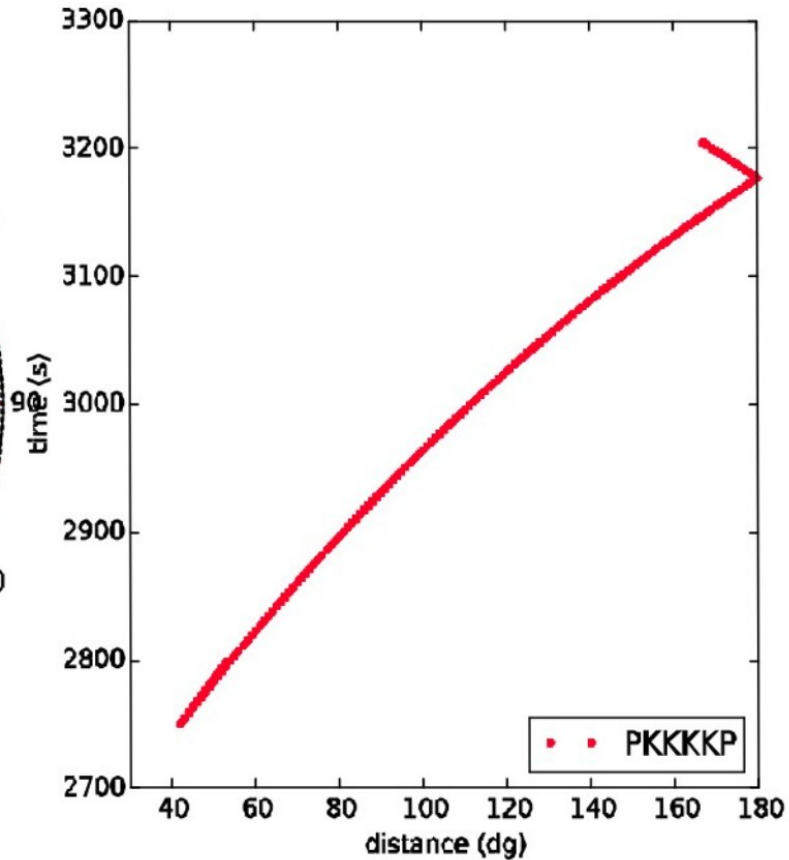
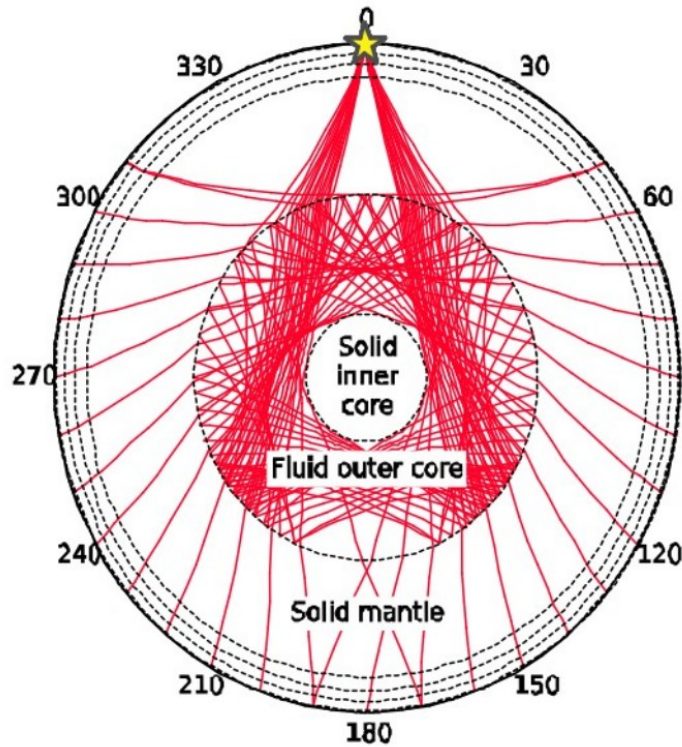
These expressions could be evaluated in the 20s and 30s using **mechanical calculating machines**

Spherical velocity structure

Global travel time observations



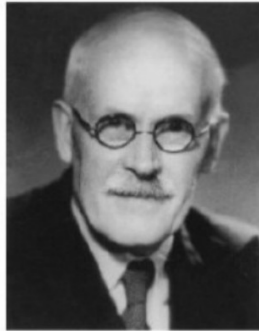
Rat theoretical predictions



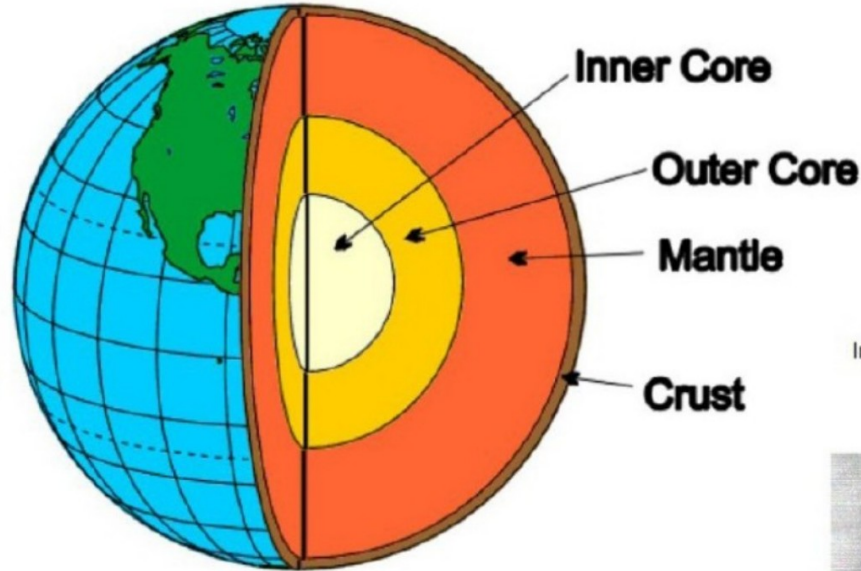
Early pioneers



Richard Oldham 1858 – 1936



Harold Jeffreys 1891 – 1989



Of all regions of the earth none invites speculation more than that which lies beneath our feet, and in none is speculation more dangerous; yet, apart from speculation, it is little that we can say regarding the constitution of the interior of the earth ... Many theories of the earth have been propounded at different times: the central substance of the earth has been supposed to be fiery, fluid, solid, and gaseous in turn, till geologists have turned in despair from the subject, and become inclined to confine their attention to the outermost crust of the earth, leaving its centre as a playground for mathematicians.

Richard Oldham 1906

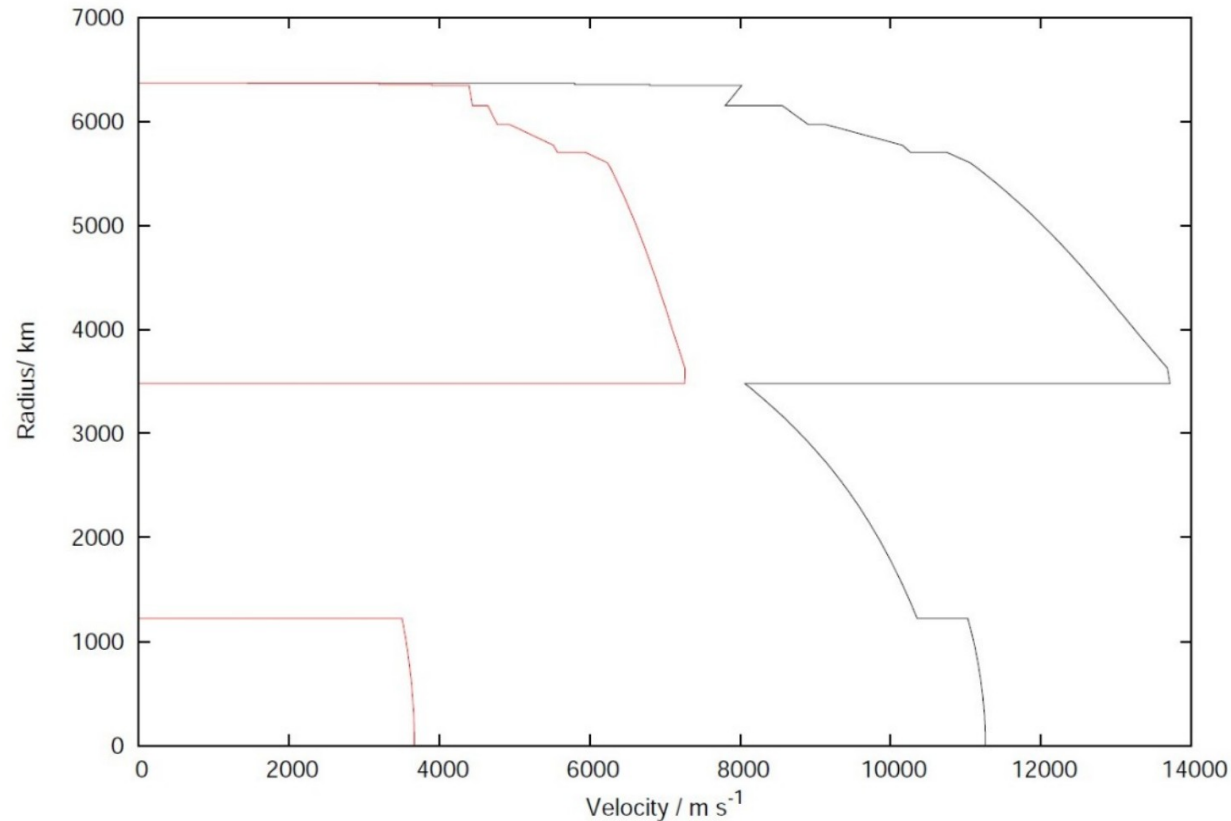


Inge Lehmann 1888 – 1993

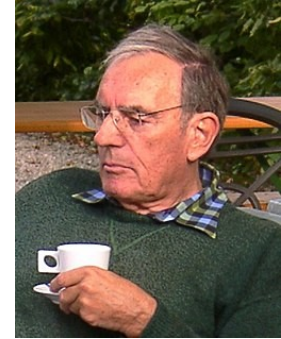


Beno Gutenberg 1889 – 1960

Spherical velocity structure



Preliminary Reference Earth Model (PREM) of Dziewonski & Anderson (1981)

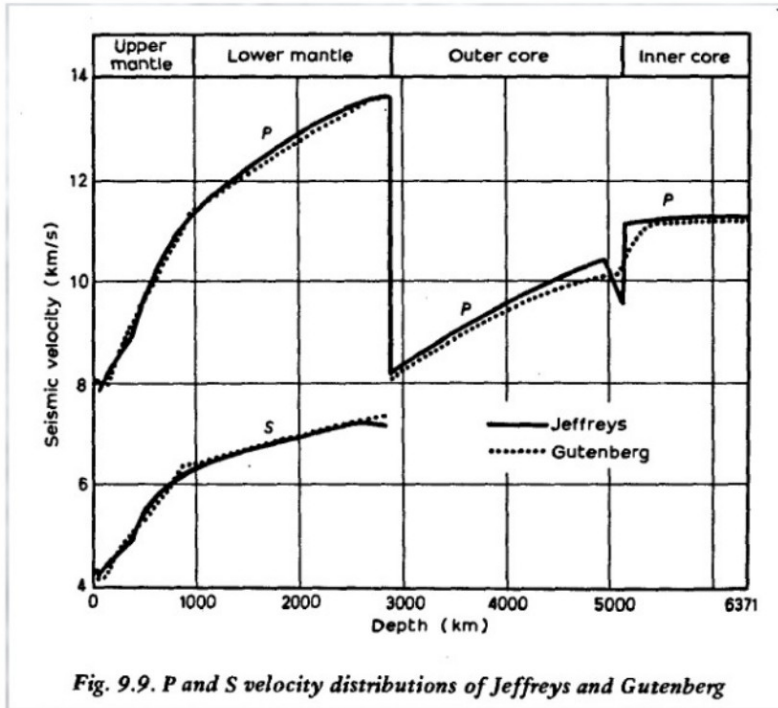


Adam Dziewonski (1936-2016)

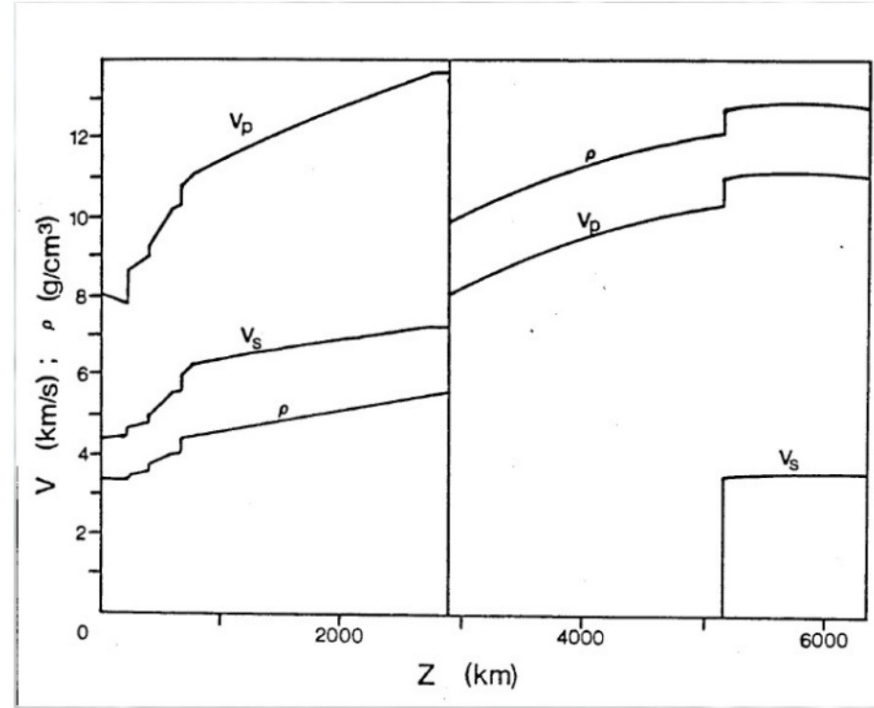


Don Anderson (1933-2014)

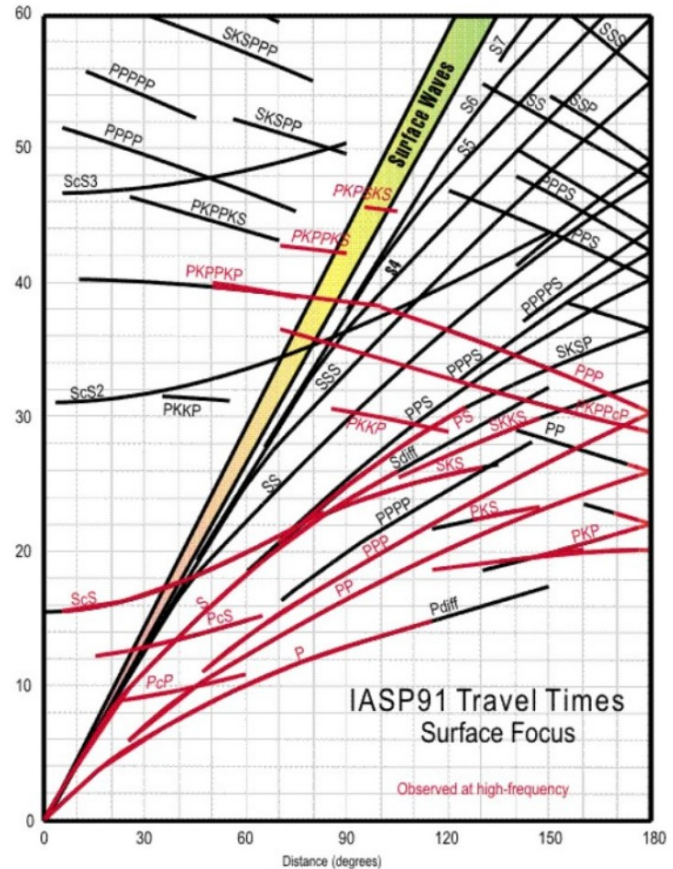
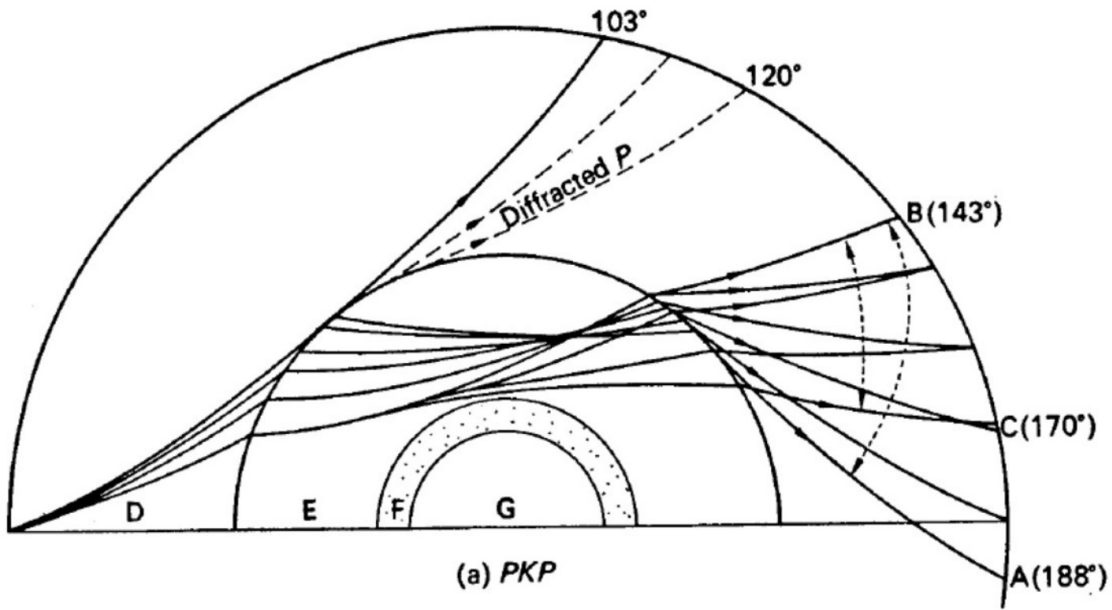
Comparison with earlier studies



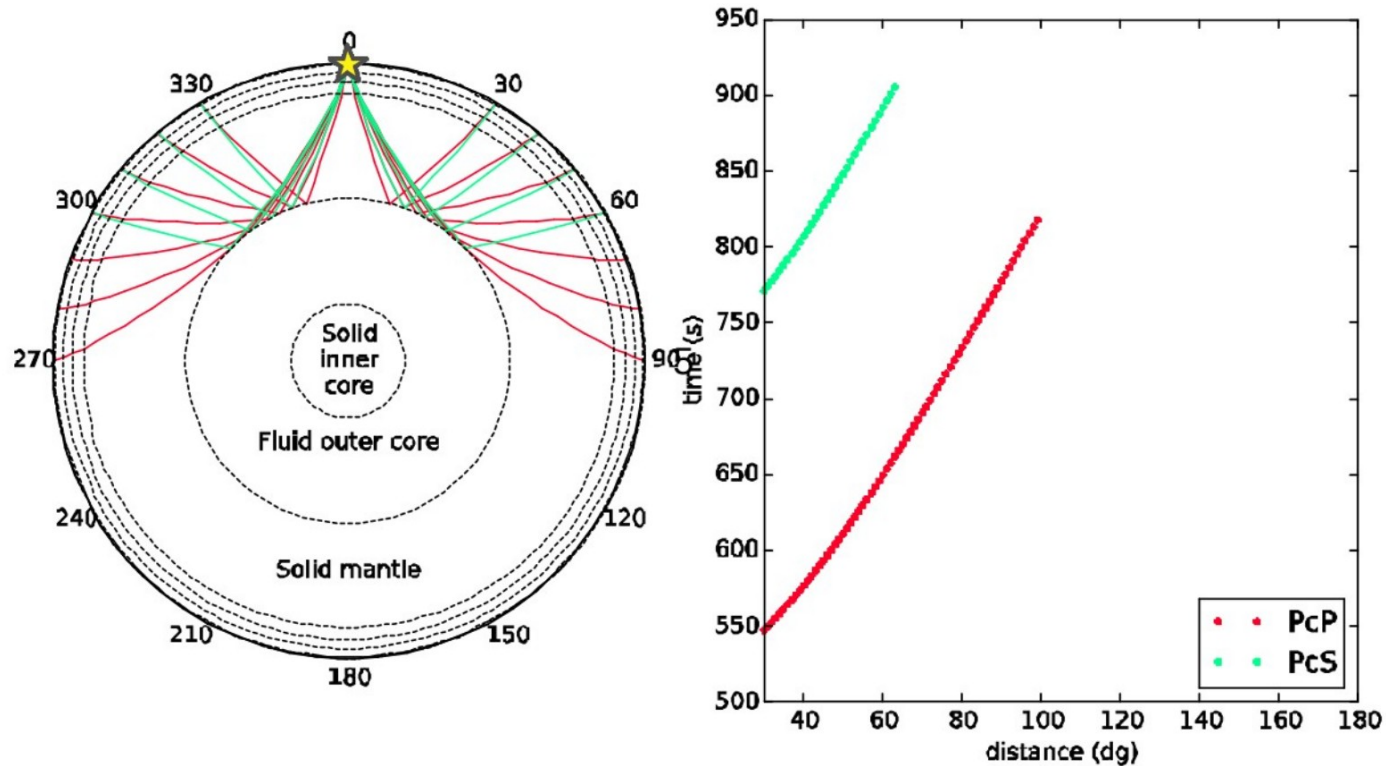
Models from the 1940s



Existence of the Earth's core

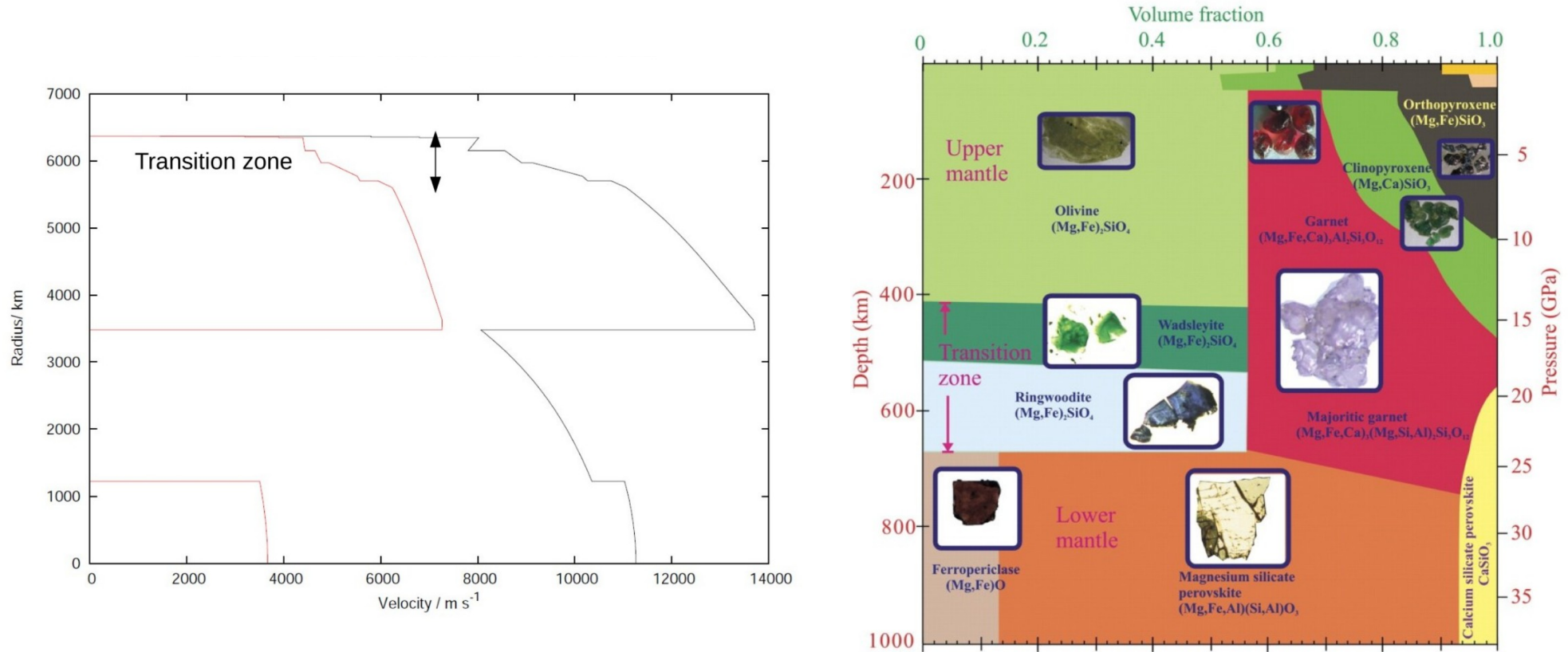


Determining internal boundaries



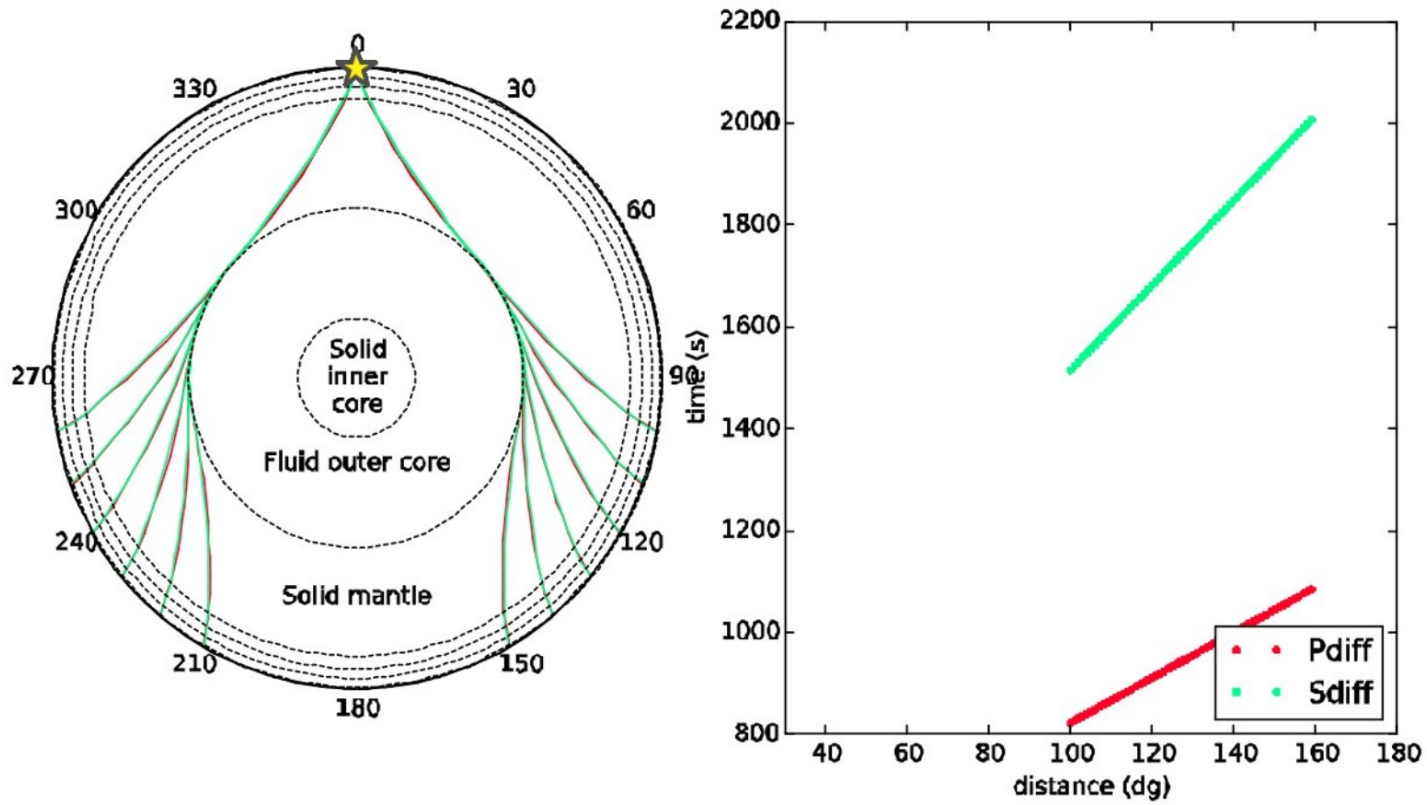
Constraints on boundary locations come largely from reflected phases

The transition zone



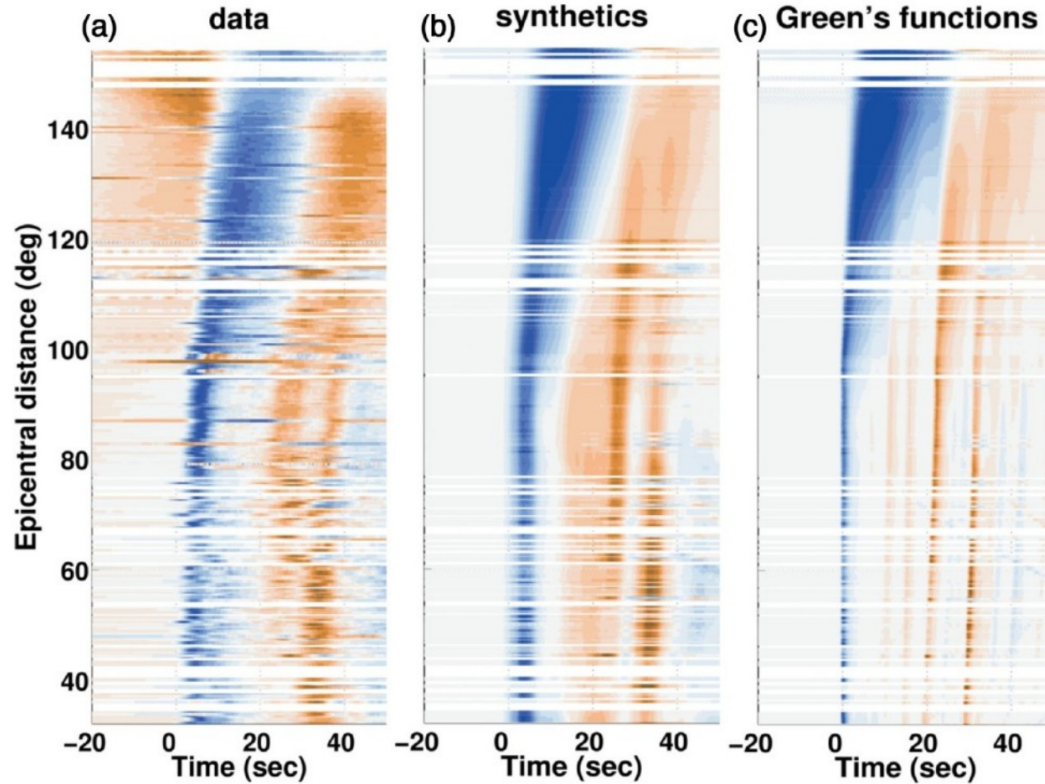
Jumps in velocity within the mid mantle believed to represent **solid-state phase changes** in silicate minerals – you don't need to know any details!

Non-geometric phases



There are also core diffracted phases that are not predicted by ray theory

Non-geometric phases



Observations and synthetics for core diffracted p-waves (normalised amplitudes)